

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: every paper reviewed in this case

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Master inventory: 5 clinical + 4 pre-clinical papers. One row per manuscript, sorted by year (newest first). Toxicities are captured per CTCAE-style term with grade, count, and rate; pre-clinical rows leave the toxicity cell blank by design.

Report	Type	Intervention	Indication / model	n	Effect size	Variance	Toxicities (type, n/N, rate)	Case fit
Mountzios/Rudin (2025) — N Engl J Med	clinical	tarlatamab (DLL3 × CD3 BiTE)	SCLC after platinum-based chemo (2L); Platinum-resistant or platinum-sensitive SCLC; ECOG 0-1	509	0.6 HR — OS (HR vs chemo)	95% CI 0.47–0.77; 95% CI; median OS 13.6 vs 8.3 mo; p<0.001	CRS (n=254, 56%); CRS G>=3 (n=254, 1%); ICANS-like neurologic event (n=254, 12%); treatment-related AE G>=3 (n=254, 24%); neutropenia G>=3 (n=254, 3%)	
Ahn/Paz-Ares (2023) — N Engl J Med	clinical	tarlatamab (DLL3 × CD3 BiTE)	previously treated SCLC (2L+); SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated	220	40 % — ORR (BICR)	95% CI 29–52; 95% CI in 10mg cohort	CRS (n=100, 51%); CRS G>=3 (n=100, 1%); ICANS-like neurologic event (n=100, 10%); treatment-related AE G>=3 (n=100, 30%); treatment discontinuation due to AE (n=100, 3%)	
Zhang/Shi (2023) — Aging (Albany NY)	preclinical	DLL3 pan-cancer expression atlas	TCGA pan-cancer transcriptomic + protein-atlas analysis	33 tumor types,	moderate — DLL3 expression detectable across 18 cancer types; co-varies with immune infiltration in subsets. Sarcoma signal present but not	vs GTEx normal-tissue baseline; translatability: low	n/a (preclinical)	weak

Yao/Pavel (2022) — Oncologist	preclinical	DLL3 as therapeutic target — review	Literature review	—	moderate — DLL3 is a 'tumor-restricted' antigen with low normal-tissue expression — attractive when present at adequate density. Surface-protein detection (e.g. SP347 IHC) is the consensus gating biomarker for DLL3-directed therapies.	vs —; translatability: med	n/a (preclinical)	weak
Italiano/Blay (2020) — Lancet Oncol	clinical	cabozantinib	advanced osteosarcoma after standard therapy (2L+); Advanced osteosarcoma (n=42) and Ewing sarcoma cohorts; ECOG 0-1; adequate organ function	42	11.9 % — ORR + 6-mo non-progression	95% CI 4.0–25.6; 95% CI; 6-mo non-progression 33.3%	hypertension G>=3 (n=42, 19%); hand-foot reaction G>=3 (n=42, 14%); hypophosphatemia G>=3 (n=42, 12%); transaminitis (ALT/AST elevation) G>=3 (n=42, 12%); thromboembolism (3/42, 7%); any treatment-related AE G>=3 (n=42, 68%)	strong
Davis/Maki (2019) — JCO	clinical	regorafenib	metastatic osteosarcoma (2L+); Metastatic osteosarcoma after at least one prior line of chemotherapy; ECOG 0-1; adequate organ function	42	3.6 months — median PFS	95% CI 2.0–7.6; vs 1.7 mo placebo; HR 0.42 (95% CI 0.21-0.85); p=0.017	hand-foot reaction G>=3 (n=22, 23%); hypertension G>=3 (n=22, 14%); fatigue G>=3 (n=22, 14%); mucositis G>=3 (n=22, 9%); any treatment-related AE G>=3 (n=22, 70%); treatment-related death G5 (0/22, 0%)	strong
Duffaud/Blay (2019) — Lancet Oncol	clinical	regorafenib	metastatic osteosarcoma (2L+); Adult metastatic osteosarcoma after conventional chemo failure; French Sarcoma Group cohort	38	65 % — non-progression at 8 weeks	95% CI 47–80; 95% CI	hand-foot reaction G>=3 (n=26, 23%); fatigue G>=3 (n=26, 15%); hypertension G>=3 (n=26, 12%); transaminitis (ALT/AST elevation) G>=3 (n=26, 8%); any treatment-related AE G>=3 (n=26, 76%); dose reduction for AE (n=26, 65%)	strong

Giffin/Coxon (2019) — Clin Cancer Res (developmental program)	preclinical	tarlatamab (DLL3 × CD3 BiTE)	DLL3-positive SCLC xenograft + humanized PBMC co-culture	n=8 in vitro n=3	strong — cytolysis demonstrated; ≥1% surface DLL3 by IHC sufficient for activity in SCLC models. Tumor regression sustained beyond treatment cessation in xenograft.	vs vehicle + PBMCs without bispecific; translatability: med	n/a (preclinical)	
Wang/Zhu (2018) — J Exp Clin Cancer Res	preclinical	Notch3/DLL3 pathway in osteosarcoma (target rationale)	Osteosarcoma cell lines (MG-63, U2OS, Saos-2); primary osteosarcoma samples	n=3 per	moderate — EGCG + doxorubicin combination suppresses Notch3/DLL3 signaling and reduces stemness in osteosarcoma; suggests pathway is operative, not just expressed.	vs vehicle + scrambled siRNA / parental cell line; translatability: low	n/a (preclinical)	weak